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List of Symbols

k_S	maximal extension of k unramified outside S	2
$G_S(k)$	$\text{Gal}(k_S/k)$	2
A'	$\text{Hom}(A, \mathcal{O}_S^\times)$	2
$\text{III}^1(k_S, T, A)$	Tate–Shafarevich Group	2
$\overline{k}_{\mathfrak{p}}$	Separable closure of $k_{\mathfrak{p}}$	2
$G_{\mathfrak{p}}$	Decomposition group of $\mathfrak{p}$	2
$H^i(K/k, A)$	$H^i(\text{Gal}(K/k), A)$	2
$k(A)$	Minimal trivializing extension of A	2
G_k	$\text{Gal}(\overline{k}/k)$	3
$\hat{H}^i(G, A)$	Tate cohomology group of A as a G -module	5
$T(K)$	Primes of K above primes in T	10
$Z_{\mathfrak{p}}$	Decomposition field of $\mathfrak{p}$	10
$G^{(p)}$	Sylow- p subgroup of G	10
$M(p)$	p -primary part of M	10

Preface

These are notes we made for a seminar class on number theoretic local-to-global principles. We would like to thank [Dr. Tim All](#) for supervising the seminar. Our primary reference for this is the book *Cohomology of Number Fields* [2]; we also reference [Kiran Kedlaya’s notes](#) [1] for class field theory and *The Arithmetic of Elliptic Curves* [3] for material on Tate–Shafarevich groups of elliptic curves.

1 The Tate–Shafarevich Group

1.1 Definition of III

Refer to [3] for more information.

Let k be a number field and S a set of (not necessarily finite) primes of k . Define k_S as the maximal extension of k unramified outside of S . Let S_∞ be the infinite places of S . Define $\mathcal{O}_{k,S}$ as

$$\mathcal{O}_{k,S} := \{a \in k : v_p(a) \geq 0 \quad \forall p \notin S \cup S_\infty\}.$$

Note that $\mathcal{O}_{k,S}$ is a localization of $\mathcal{O}_k$. Define

$$\mathbf{N}(S) := \mathbf{N} \cap \mathcal{O}_{k,S}^\times.$$

Let $G_S(k) := \text{Gal}(k_S/k)$. If A is a $G_S(k)$ -module with $|A| \in \mathbf{N}(S)$, define $A' := \text{Hom}(A, \mathcal{O}_S^\times)$ where $\mathcal{O}_S = \mathcal{O}_{k_S,S}$.

We can now define the Tate–Shafarevich group. Let $T \subseteq S$ be a set of primes. We define III^1 via the following exact sequence:

$$0 \longrightarrow \text{III}^1(k_S, T, A) \longrightarrow H^1(k_S/k, A) \xrightarrow{\text{res}} \prod_{p \in T} H^1(\overline{k_p}/k_p, A)$$

The map from $H^1(k_S/k, A)$ is a componentwise restriction, which we will now go into detail about. In general, we have the following: let k be a number field and K any Galois extension of k . For a prime p in k , we have a maximal ideal $\mathfrak{p}$ in K lying over p . We then have the decomposition group

$$G_{\mathfrak{p}} = \text{Gal}(K_{\mathfrak{p}}/k_p).$$

We can then consider the restriction map $\text{res} : H^i(K/k, A) \rightarrow H^i(G_{\mathfrak{p}}, A)$. While this appears to depend on the choice of prime lying over p , it turns out that the choice is irrelevant. If $\mathfrak{p}_1, \mathfrak{p}_2$ both lie over p , then we have an automorphism $\sigma \in \text{Gal}(K/k)$ such that $\sigma G_{\mathfrak{p}_1} \sigma^{-1} = G_{\mathfrak{p}_2}$. Since inner automorphisms act trivially on group cohomology, we obtain *canonical* isomorphisms $H^i(G_{\mathfrak{p}_1}, A) \simeq H^i(G_{\mathfrak{p}_2}, A)$, and thus the choice of $\mathfrak{p}$ does not matter.

Now, we have a surjection

$$\text{Gal}(\overline{k_p}/k_p) \twoheadrightarrow \text{Gal}(K_{\mathfrak{p}}/k_p).$$

This gives an action of $\text{Gal}(\overline{k_p}/k_p)$ on A . Since $G_{\mathfrak{p}} = \text{Gal}(K_{\mathfrak{p}}/k_p)$ is a quotient of $\text{Gal}(\overline{k_p}/k_p)$, we have the following chain of maps

$$H^i(K/k, A) \xrightarrow{\text{res}} H^i(G_{\mathfrak{p}}, A) \xrightarrow{\text{inf}} H^i(\overline{k_p}/k_p, A).$$

For the case of $K = k_S$, Proposition (8.6.1) of [2] states that in fact $K_p = \overline{k_p}$, so the inflation map is from $G/1$ to G and is thus unnecessary.

Define $k(A)$ to be the minimal extension of k such that $\text{Gal}(k_S/k(A))$ acts trivially on A . Then for some subextension $k_S/K/k(A)$, we can define the relative Tate–Shafarevich group

$$0 \longrightarrow \text{III}^1(K/k, T, A) \longrightarrow H^1(K/k, A) \xrightarrow{\text{res}} \prod_{p \in T} H^1(K_p/k_p, A)$$

For example, let A be trivial as an abelian group. Then $H^1(k_S/k, A) = 0$ so since $\text{III}^1(k_S, T, A) \subseteq H^1(k_S/k, A)$, we obtain $\text{III}^1(k_S, T, A) = 0$.

If $S = \emptyset$ and $k = \mathbf{Q}$, then a theorem of Minkowski tells us $\mathbf{Q}_S = \mathbf{Q}$ so $G_S(k)$ is trivial and all Tate–Shafarevich groups vanish because cohomology vanishes.

If S is any set of primes of k , then $k_S^\times$ is a $G_S(k)$ -module. By Hilbert 90, we obtain $H^1(k_S/k, k_S^\times) = 0$ so $\text{III}^1(k_S, T, A) = 0$ for all $T \subseteq S$.

For the case of an abelian variety A/K (such as an elliptic curve), let S be the set of all places of K . Then the usual Tate–Shafarevich group of A is defined as

$$\text{III}(A/K) = \text{III}^1(K_S, S, A(\bar{K})).$$

We would be remiss to not mention the following:

Conjecture 1: (Tate–Shafarevich)

$\text{III}(A/K)$ is finite for all abelian varieties A and number fields K .

Conjecture 2: (Birch–Swinnerton-Dyer)

Let $E/\mathbf{Q}$ be an elliptic curve, $L(E, s)$ its Hasse–Weil L -function, n its conductor, and r its algebraic rank. Then

$$\frac{L^{(r)}(E, 1)}{r!} = \frac{|\text{III}(E/\mathbf{Q})| w_\infty R_E \prod_{p|n} w_p}{|E(K)_{\text{tors}}|^2}$$

where w_p are the Tamagawa numbers, R_E is the regulator of E , w_∞ is a multiple of the real period of E , and $E(K)_{\text{tors}}$ is the torsion subgroup of $E(K)$.

1.2 Motivating the definition of III

While the definition for III clearly talks about a relation between local and global, it's not clear what exactly that relation is; we aim to explain this relation here. We begin by recalling a classical local-to-global result in number theory, which we discuss in detail [here](#):

Theorem: Hasse–Minkowski

Let Q be a quadratic form over a global field k ($\text{char } k \neq 2$). Then Q represents 0 nontrivially over k (i.e. $Q(\mathbf{x}) = 0$ for some $\mathbf{x} \neq \mathbf{0}$) if and only if Q is *isotropic* (represents 0 nontrivially) over all completions k_ν of k .

In a more geometric language, we might say that the variety defined by the zero set of Q has a k -rational point if and only if it has k_ν -rational points for each place ν of k . In the language of Tate–Shafarevich groups, we might be tempted to say something like $\text{III}(k, Q) = 0$ (see the last section of [these notes](#) for more about this); however, the varieties defined by Q are in general not abelian varieties. That is, these varieties do not necessarily have a meaningful group structure on which we can easily define a III . As it so happens, elliptic curves (and more generally abelian varieties) *do* have a nice group structure on which we can meaningfully define III ; from these we get the classical form of the Tate–Shafarevich group.

First we motivate the usage of cohomology. Fix a number field k and let S be the set of all places of k so that $G := G_S(k) = G_k = \text{Gal}(\bar{k}/k)$, and let $A = A(\bar{k})$ be an abelian variety. We note that

$$A(k) = A(\bar{k})^{G_k} = A^{G_k} = H^0(G, A).$$

This motivates the introduction of group cohomology into the study of $A(k)$. Let $\phi : A \rightarrow A$ be a surjective G -equivariant endomorphism of modules, and denote by $A[\phi]$ the kernel of ϕ . We have the short exact sequence

$$0 \longrightarrow A[\phi] \longrightarrow A \xrightarrow{\phi} A \longrightarrow 0$$

Taking G -cohomology, we obtain the cohomology long-exact sequence

$$0 \longrightarrow A[\phi]^G \longrightarrow A^G \xrightarrow{\phi} A^G \longrightarrow H^1(G, A[\phi]) \longrightarrow H^1(G, A) \xrightarrow{\phi} H^1(G, A) \longrightarrow \dots$$

Looking at exactness at $H^1(G, A[\phi])$, we obtain the following short exact sequence

$$0 \longrightarrow A^G/\phi(A^G) \xrightarrow{\delta} H^1(G, A[\phi]) \longrightarrow H^1(G, A)[\phi] \longrightarrow 0$$

In the proof of the Mordell–Weil theorem, a crucial step is 2-descent (or, in our case, ϕ -descent) which is showing $A^G/\phi(A^G) = A(k)/\phi A(k)$ is finite. The above exact sequence shows that it suffices to prove that $H^1(G, A[\phi])$ is finite. This ends up being true: It turns out that this is a group of “homogeneous spaces,” which have the structure of algebraic curves, and that this group is (comparatively) easy to understand. We know that we have an injective map $\delta : A^G/\phi(A^G) \rightarrow H^1(G, A[\phi])$, so it suffices to study the cokernel of δ to understand $A^G/\phi(A^G)$.

It turns out that elements in the image of δ correspond to the homogeneous spaces that happen to have a k -rational point. If we do the same argument locally, we obtain the exact sequence.

$$0 \longrightarrow A^{G_\nu}/\phi(A^{G_\nu}) \xrightarrow{\delta} H^1(G_\nu, A[\phi]) \longrightarrow H^1(G_\nu, A)[\phi] \longrightarrow 0$$

Where G_ν is the decomposition group of ν in G . In this case, elements in the image of δ correspond to k_ν -points of the same algebraic curves. We can consider the following commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & A^G/\phi(A^G) & \xrightarrow{\delta} & H^1(G, A[\phi]) & \longrightarrow & H^1(G, A)[\phi] \longrightarrow 0 \\ & & \downarrow \text{res} & & \downarrow \text{res} & & \downarrow \text{res} \\ 0 & \longrightarrow & \prod_\nu A^{G_\nu}/\phi(A^{G_\nu}) & \xrightarrow{\delta} & \prod_\nu H^1(G_\nu, A[\phi]) & \longrightarrow & \prod_\nu H^1(G_\nu, A)[\phi] \longrightarrow 0 \end{array}$$

The image of δ in the bottom exact sequence is classified by collections of homogeneous spaces C_ν that all have k_ν -rational points, immediately, we see that an element of $\prod_\nu H^1(G_\nu, A[\phi])$ can only come from $A^G/\phi(A^G)$ if it is “diagonal” in the sense that $C_\nu = C$ for all ν , and moreover $C = C_\nu$ has a k_ν point for all ν . This second condition about the existence of k_ν points is equivalent to being in the image of the lower δ .

The upshot is that the “diagonal” elements of $\prod_\nu H^1(G_\nu, A)[\phi]$ and the elements of $H^1(G, A)[\phi]$ measure, respectively, the homogeneous spaces that do not have k_ν -points for all ν , and the homogeneous spaces that do not have k -rational points. So the kernel of the map

$$\text{res} : H^1(G, A)[\phi] \rightarrow \prod_\nu H^1(G_\nu, A)[\phi]$$

measures the homogeneous spaces that don’t have k -rational points but do have k_ν points for all ν . We then drop the ϕ dependence and obtain the definition

$$\text{III}(G, A) := \bigcap_\nu \ker(H^1(G, A) \rightarrow H^1(G_\nu, A))$$

2 Local-to-Global Results

2.1 Some Cohomology

Our goal for moving forward will be to compute $\text{III}^1(k_S, T, A)$ for various $G_k(S)$ -modules A ; we will be especially interested in the case of μ_m , which is one of the most natural $G_k(S)$ modules.

Our first step to doing this will be the following lemma (9.1.4 in [2]).

Lemma 1

Let p be a prime, $m \geq 1$ a natural number and $G \subseteq (\mathbf{Z}/p^m\mathbf{Z})^\times = \text{Aut}(\mathbf{Z}/p^m\mathbf{Z})$ act on $A = \mathbf{Z}/p^m\mathbf{Z}$ in the obvious way. Then we have

$$\hat{H}^i(G, A) \cong \begin{cases} \mathbf{Z}/2\mathbf{Z} & \text{if } p = 2, m \geq 2, -1 \in G \\ 0 & \text{otherwise} \end{cases}.$$

Proof. We begin with the case $p > 2$. Let $G_1 = G \cap \ker((\mathbf{Z}/p^m\mathbf{Z})^\times \rightarrow (\mathbf{Z}/p\mathbf{Z})^\times)$ be the Sylow- p subgroup of G . Since G is abelian, G_1 is a normal subgroup of G and we have the inflation-restriction exact sequence

$$0 \longrightarrow H^1(G/G_1, A^{G_1}) \xrightarrow{\text{inf}} H^1(G, A) \xrightarrow{\text{res}} H^1(G_1, A)^{G/G_1} \xrightarrow{\text{trans}} H^2(G/G_1, A^{G_1}) \quad (1)$$

We claim that the ends of this exact sequence vanish.

In fact, $H^i(G/G_1, A^{G_1}) = 0$ for all $i \geq 1$. This is because cohomology groups $H^i(N, B)$ are always $|N|$ torsion, and likewise if B is n -torsion then $H^i(N, B)$ is also n -torsion. In our case, A^{G_1} is p^m -torsion since it is a subgroup of $\mathbf{Z}/p^m\mathbf{Z}$, and $|G/G_1| = p - 1$, so $H^i(G/G_1, A^{G_1})$ is $\gcd(p^m, p - 1) = 1$ -torsion and therefore is trivial. Substituting this into equation (1), we get the exactness of

$$0 \xrightarrow{\text{inf}} H^1(G, A) \xrightarrow{\text{res}} H^1(G_1, A)^{G/G_1} \xrightarrow{\text{trans}} 0$$

so $H^1(G, A) \simeq H^1(G_1, A)^{G/G_1}$. We wish to show that $H^1(G, A) = 0$, which we will do by showing that $H^1(G_1, A) = 0$; we can do that by showing that $\hat{H}^0(G_1, A) = 0$ since A is finite and has Herbrand quotient 1. We know that $\hat{H}^0(G_1, A) = A^{G_1}/N_{G_1}A$. Let $G_1 = \langle \alpha \rangle$ with $\alpha = 1 + p^s u$. Then by lifting the exponent we have $|\langle G \rangle| = p^{m-s}$. For A^G , we have

$$\alpha x \equiv x \pmod{p^m} \implies p^m | (\alpha - 1)x \implies p^m | p^s u x$$

So we have that $A^{G_1} = p^{m-s}\mathbf{Z}/p^m\mathbf{Z}$. Similarly, we have

$$N_{G_1}(x) = \sum_{k=0}^{p^{m-s}-1} \alpha^k x = \frac{\alpha^{p^{m-s}} - 1}{\alpha - 1} x = \gamma x.$$

By lifting the exponent, $v_p(\alpha^{p^{m-s}} - 1) = m$, so $v_p(\gamma) = p^{m-s}$. Therefore, $\text{im}(N_{G_1}) = p^{m-s}\mathbf{Z}/p^m\mathbf{Z}$, so $\hat{H}^0(G_1, A) = 0$. Since the Herbrand quotient is 1, we have $\hat{H}^1(G_1, A) = 0$ and therefore $\hat{H}^1(G, A) = 0$. Since A is finite and G is cyclic, we obtain $\hat{H}^i(G, A) = 0$ for all i .

Now let $p = 2$. If $m = 1$, then $G \subseteq (\mathbf{Z}/2\mathbf{Z})^\times = 1$ is trivial, so the cohomology groups $\hat{H}^i(G, A)$ vanish, so we assume $m \geq 2$. Let $U_i = \ker(\mathbf{Z}_p^\times \rightarrow (\mathbf{Z}/2^i\mathbf{Z})^\times)$. With this, we can consider three difference cases for the structure of $G \subseteq (\mathbf{Z}/2^m\mathbf{Z})^\times$.

- $G = \{\pm 1\} \times \langle \alpha \rangle$.
- $G = \langle \alpha \rangle$ when $\alpha \in U_2$.
- $G = \langle \alpha \rangle$ when $-\alpha \in U_2$.

Note that these are all possible cases: if G is not cyclic, then $-1 \in G$ and it is generated by -1 and another element. If G is cyclic, then it is generated by some α and either α or $-\alpha$ is in U_2 .

In case (b.) the same argument as the odd prime case works, however we do not even need to pass the the Sylow- p subgroup since G is already a p -group. For case (c.), let $-\alpha \in U_i - U_{i+1}$, then by lifting the exponent we have (for $k > 1$) $v_2(\alpha^{2^k} - 1) = v_2((-\alpha)^{2^k} - 1) = 2^{s+k}$, so $|\langle \alpha \rangle| = |2^{m-s}|$. Then by the same techniques as the odd p case we obtain

$$A^G = \ker(\alpha - 1 : A \rightarrow A) = 2\mathbf{Z}/2^m\mathbf{Z} \quad N_G(A) = 2\mathbf{Z}/2^m\mathbf{Z}$$

so we have $\hat{H}^0(G, A) = 0$ and since the Herbrand quotient is 1 we have $\hat{H}^i(G, A) = 0$. WLOG let $\alpha \in U_2$ (note that if $\alpha \notin U_2$, we can select $-\alpha$ as the generator for $G/\{\pm 1\}$.) Then we have the inflation-restriction exact sequence on $\{\alpha\} \subseteq G$.

$$0 \longrightarrow H^i(G/\langle \alpha \rangle, A^{(\alpha)}) \xrightarrow{\inf} H^i(G, A) \xrightarrow{\text{res}} H^i(\langle \alpha \rangle, A)$$

Note that the exact sequence exists for all i because $H^i(\langle \alpha \rangle, A) = 0$ for all $i > 0$ by part (b). Therefore we obtain the exact sequence

$$0 \rightarrow H^i(G/\langle \alpha \rangle, A^{(\alpha)}) \rightarrow H^i(G, A) \rightarrow 0$$

Computing $\hat{H}^1(G/\langle \alpha \rangle, A^{(\alpha)})$, we see that $(A^{(\alpha)})^{G/\langle \alpha \rangle} = 2^{m-1}\mathbf{Z}/2^m\mathbf{Z}$, and $N_{G/\langle \alpha \rangle}$ is the zero map, so $\hat{H}^1(G/\langle \alpha \rangle, A^{(\alpha)}) = \mathbf{Z}/2\mathbf{Z}$, and since the Herbrand quotient is 1 we have $H^i(G/\langle \alpha \rangle, A^{(\alpha)}) = \mathbf{Z}/2\mathbf{Z}$, so by the exact sequence, $H^i(G, A) = \mathbf{Z}/2\mathbf{Z}$.

In the above, we did not consider negative cohomology. The authors are aware of two methods that can be used here, the first is to use a homological version of inflation-restriction (corestriction-coinflation) and repeat the previous arguments in this context.

The second method goes as follows: by Theorem 2.6.9 of [2], for a profinite group G and compact G -module A , there are isomorphisms

$$H_i(G, A)^\vee \cong H^i(G, A^\vee)$$

We can use this in our situation. In our case, $A^\vee$ is generated by $\phi : A \rightarrow \mathbf{Q}/\mathbf{Z}$ that takes 1 to $1/2^m$. $A^\vee$ is abstractly isomorphic to $\mathbf{Z}/2^m\mathbf{Z}$ but has the opposite G -action, as $(g\phi)(x) = \phi(g^{-1}x) = g^{-1} \cdot \phi(x)$, where $g^{-1} \cdot \phi$ is literal multiplication, so we get an isomorphism $A^\vee \cong A^{op}$, where A^{op} is A with the opposite G action. Since G is abelian, the map $g \mapsto g^{-1}$ is an automorphism of G , so the cohomology of A and A^{op} is identical and we get isomorphisms

$$H_i(G, A) \cong H_i(G, A)^\vee \cong H^i(G, A^\vee) \cong H^i(G, A^{op}) \cong H^i(G, A) \cong \mathbf{Z}/2\mathbf{Z}.$$

We now use this lemma to prove a fact about Galois cohomology:

The Special Case

Let k be a field, m a positive integer coprime to $\text{char}(k)$. We say we are in the *special case* (k, m) if $r = \nu_2(m) \geq 2$ and $-1 \in \text{im}(\chi_{\text{cycl}})$ where χ_{cycl} is the cyclotomic character

$$\chi_{\text{cycl}} : \text{Gal}(k(\mu_{2^r})/k) \rightarrow (\mathbf{Z}/2^r\mathbf{Z})^\times$$

which gives the action of $\text{Gal}(k(\mu_{2^r})/k)$ on μ_{2^r} .

Proposition 1

Let p be a prime, $r \in \mathbf{N}$, k any field with $\text{char}(k) \neq p$. Then

$$\hat{H}^i(\text{Gal}(k(\mu_{p^r})/k), \mu_{p^r}) = 0 \quad \text{for all } i \in \mathbf{Z}$$

except when $p = 2$, $r \geq 2$, and we are in the *special case* $(k, 2^r)$. Then

$$\hat{H}^i(\text{Gal}(k(\mu_{p^r})/k), \mu_{p^r}) \cong \mathbf{Z}/2\mathbf{Z} \quad \text{for all } i \in \mathbf{Z}.$$

Let $p = 2$ and $r \geq 2$; the *special case* occurs **if and only if**

- ✦ $\text{char}(k) = 0$ and $\mathbf{Q}(\mu_{2^r}) \cap k$ is real **OR**
- ✦ $\text{char}(k) = l \equiv -1 \pmod{2^r}$ and $\mathbf{F}_l(\mu_{2^r}) \cap k = \mathbf{F}_l$

Exercise 1

In positive characteristic, the *special case* occurs **if and only if** $k(\mu_{2^{r+1}}) = k(\mu_{2^r})$ and $[k(\mu_{2^r}) : k] = 2$. The proof of this is left as an exercise to the reader.

Proof. $G = \text{Gal}(k(\mu_{p^r})/k)$ acts on μ_{p^r} via the cyclotomic character $\chi_{\text{cycl}} : G \hookrightarrow (\mathbf{Z}/p^r\mathbf{Z})^\times$; we can apply **lemma 1**, so it suffices to identify the *special cases*.

If $\text{char}(k) = 0$, then χ_{cycl} factors

$$\chi_{\text{cycl}} : G \xrightarrow{\sim} \text{Gal}(\mathbf{Q}(\mu_{2^r})/k \cap \mathbf{Q}(\mu_{2^r})) \hookrightarrow (\mathbf{Z}/2^r\mathbf{Z})^\times.$$

Since $-1 \in (\mathbf{Z}/2^r\mathbf{Z})^\times \cong \text{Gal}(\mathbf{Q}(\mu_{2^r})/\mathbf{Q})$ corresponds to complex conjugation, it follows that $-1 \in \text{im}(\chi_{\text{cycl}})$ if and only if $k \cap \mathbf{Q}(\mu_{2^r})$ is real.

Now let $\text{char}(k) = l \notin \{0, 2\}$. Then χ_{cycl} factors

$$\chi_{\text{cycl}} : G \xrightarrow{\sim} \text{Gal}(\mathbf{F}_l(\mu_{2^r})/k \cap \mathbf{F}_l(\mu_{2^r})) \hookrightarrow (\mathbf{Z}/2^r\mathbf{Z})^\times$$

$\text{Gal}(\mathbf{F}_l(\mu_{2^r})/\mathbf{F}_l(\mu_{2^r}) \cap k)$ is cyclic generated by the Frobenius σ , where

$$\chi_{\text{cycl}}(\sigma) = l^{[\mathbf{F}_l(\mu_{2^r}) \cap k : \mathbf{F}_l]}.$$

It follows that $-1 \in \text{im}(\chi_{\text{cycl}})$ if and only if $l^{[\mathbf{F}_l(\mu_{2^r}) \cap k : \mathbf{F}_l]} \equiv -1 \pmod{2^r}$ for some n . Since $(\mathbf{Z}/2^r\mathbf{Z})^\times$ decomposes as

$$(\mathbf{Z}/2^r\mathbf{Z})^\times \cong \{\pm 1\} \times \mathbf{Z}/2^{r-2}\mathbf{Z},$$

we can write $l^{[\mathbf{F}_l(\mu_{2^r}) \cap k : \mathbf{F}_l]} = -1 \cdot g$ for some $g \in \mathbf{Z}/2^{r-2}\mathbf{Z}$. In order to preserve the -1 factor, n must be odd. However, $\mathbf{Z}/2^{r-2}\mathbf{Z}$ only has 2-torsion; it follows that $g = 1$ and thus $l^{[\mathbf{F}_l(\mu_{2^r}) \cap k : \mathbf{F}_l]} \equiv -1 \pmod{2^r}$. The same argument again shows that we must have $[\mathbf{F}_l(\mu_{2^r}) \cap k : \mathbf{F}_l] = 1$ and $l \equiv -1 \pmod{2^r}$. ■

Another Special Case

Let k be a global field, $m \in \mathbf{N}$ coprime to $\text{char}(k)$, $r := \nu_2(m)$, and T a set of primes of k . Then we say we are in the *special case* (k, m, T) if we are in the *special case* (k, m) and all primes $\mathfrak{p} \in T$ split in $k(\mu_{2^r})/k$ (more than one prime of $k(\mu_{2^r})$ lies above $\mathfrak{p}$).

Lemma 2

Let k be a global field with $\text{char}(k) \neq 2$. Suppose $k(\mu_{2^r})/k$ is cyclic and $\delta(T) > 1/2$ (where δ denotes Dirichlet density, assuming it exists). Then we are not in the *special case* $(k, 2^r, T)$.

Proof. If we are in the *special case* $(k, 2^r)$ and $k(\mu_{2^r})/k$ is cyclic, we must have $[k(\mu_{2^r}) : k] = 2$. The Chebotarev Density Theorem implies that the primes which do not split have density $1/2$, and must then have nontrivial intersection with T . ■

Example:

1. The extension $k(\mu_{2^r})/k$ can be non-cyclic only if k is a number field, $r \geq 3$, and $k \cap \mathbf{Q}(\mu_{2^r})$ is real.
2. If $k(\mu_{2^r})/k$ is not cyclic, then all primes $\mathfrak{p} \nmid 2$ split in $k(\mu_{2^r})/k$. These primes are unramified and therefore have a cyclic decomposition group.
3. If $\delta(T) > 1/2$, then by the previous two remarks and **lemma 2**, the *special case* (k, m, T) is equivalent to the statement
 k is a number field, $m \in \mathbf{N}$, $r = \nu_2(m) \geq 3$, $k(\mu_{2^r})/k$ is not cyclic, and all primes $\mathfrak{p}$ dividing 2 in T split in

$$k(\mu_{2^r})/k.$$

It may be interesting to see an immediate number-theoretic application of these cohomology computations. Indeed such a result exists.

Proposition 2

Let k be a global field and m not divisible by $\text{char}(k)$. Then

$$k(\mu_m)^{\times m} \cap k^\times = k^{\times m}$$

unless we are in the special case (k, m) , in which case we have

$$(k(\mu_m)^{\times m} \cap k^\times)/k^{\times m} \cong \mathbf{Z}/2\mathbf{Z}.$$

Proof. Consider the “Kummer exact sequence:”

$$1 \longrightarrow \mu_m \longrightarrow k(\mu_m)^\times \xrightarrow{\cdot m} k(\mu_m)^\times \longrightarrow 1$$

Taking cohomology and applying “theorem 90”, we obtain

$$1 \longrightarrow \mu_m \cap k \longrightarrow k^\times \xrightarrow{\cdot m} k(\mu_m)^{\times m} \cap k \longrightarrow H^1(\mu_m) \longrightarrow 1$$

Exactness yields $H^1(k, \mu_m) = (k(\mu_m)^{\times m} \cap k)/k^{\times m}$, and the result follows. ■

2.2 Local-to-global for m^{th} powers

Recall the following theorem from “elementary” number theory:

Theorem 1

let $x \in \mathbf{Q}$. Then x is an m^{th} power if and only if it is an m^{th} power in $\mathbf{Q}_p$ for all $p \leq \infty$.

The proof of this is relatively simple: write $x = \pm p_1^{v_{p_1}(x)} \dots p_n^{v_{p_n}(x)}$, then the condition for $p = \infty$ implies that we have a valid sign, and the condition for finite p implies $v_p(x)$ is a multiple of m . We can then explicitly describe the m^{th} root.

This theorem is nice, but we would like it to be generalized in two ways. Firstly, and most obviously, we would like to generalize $\mathbf{Q}$ to an arbitrary number field. Secondly, and more subtly, we would like to loosen the requirement from “all primes” to “most primes” for a sufficiently broad definition of most. A theorem of this form will be the fruit of our cohomological labor. We will do this with the power of III.

We now need to define some notation. Let S and T be sets of primes of k . Let $\delta(S)$ denote the Dirichlet density of S , we say that $T \subsetneq S$ if $\delta(T - S) = 0$. For a Galois extension of number fields K/k , denote by $\text{cs}(K/k)$ the set of primes of k that split completely in K .

Theorem 2

Let k be a global field, S a set of primes of k , $T \subseteq S$, A be a finite $G_S(k)$ -module. Then $\text{III}^1(k_S, T, A) = 0$ in the following cases:

- (i) A has trivial $G_S(k)$ -action and $\delta(T) > 1/p$, where p is the smallest prime dividing $|A|$.
- (ii) $A = \mu_m$ with $m = p_1^{r_1} \dots p_n^{r_n}$ for $p_i \in \mathbf{N}(S)$ and

$$\delta(\text{cs}(k(\mu_{p_i}^{r_i})/k) \cap T) > \frac{1}{p_i[k(\mu_{p_i}^{r_i}) : k]}$$

- for all $i = 1, \dots, n$, except when we are in the special case (k, m, S) where $\text{III}^1(k_S, T, \mu_m) = \mathbf{Z}/2\mathbf{Z}$.
- (iii) $\text{cs}(k(A)/k) \subsetneq T$ and $|\text{Gal}(k(A)/k)| = \text{lcm}\{|\text{Gal}(k_{\mathfrak{p}}(A)/k_{\mathfrak{p}})| : \mathfrak{p} \in T\}$.

Proof. We need the following homological lemma:

Lemma 3: Additivity of Group Cohomology

Let A, B be G -modules. Then

$$H^i(G, A \oplus B) = H^i(G, A) \oplus H^i(G, B).$$

Proof. Recall the standard resolution $X^n(A) = \text{Map}(G^{n+1}, A)$, where $\text{Map}(X, A)$ is the group of continuous functions from X to A . The universal property of the product shows that

$$\text{Map}(X, A \oplus B) = \text{Map}(X, A \times B) = \text{Map}(X, A) \times \text{Map}(X, B) = \text{Map}(X, A) \oplus \text{Map}(X, B).$$

The result follows by tracking through the construction of the cohomology groups $H^i(G, A \oplus B)$ using the resolution $X^n = \text{Map}(G^{n+1}, A) \oplus \text{Map}(G^{n+1}, B)$. ■

Applying additivity to the definition of III , we have III^i is the kernel of res^i ; it follows that $\text{III}^i(k_S, T, A \oplus B) = \text{III}^i(k_S, T, A) \oplus \text{III}^i(k_S, T, B)$. We will prove that (i) holds for the case $A = \mathbf{Z}/p^r\mathbf{Z}$; additivity of III^1 proves the result for any A .

For $A = \mathbf{Z}/p^r\mathbf{Z}$ a trivial $G_S(k)$ -module, we want to show injectivity of the morphism

$$\text{res}^1 : \text{Hom}_{\text{cts}}(\text{Gal}(k_S/k), \mathbf{Z}/p^r\mathbf{Z}) \cong H^1(k_S/k, \mathbf{Z}/p^r\mathbf{Z}) \rightarrow \prod_{\mathfrak{p} \in T} H^1(k_{\mathfrak{p}}, \mathbf{Z}/p^r\mathbf{Z}).$$

If φ is in this kernel, then $\ker(\varphi)$ corresponds to a cyclic extension L/k of degree p^s ($0 \leq s \leq r$) which is unramified outside S and completely split at T . By assumption $\delta(T) > 1/p$, so Chebotarev Density Theorem implies that $s = 0$ and thus $\varphi = 0$.

For (ii), recall that $\mu_m = \mathbf{Z}/m\mathbf{Z}$; the same argument as in the proof of (i) allows us to reduce to the case where

$m = p^r$. Let $K := k(A) = k(\mu_{p^r})$. We have the commutative exact diagram

$$\begin{array}{ccccccc}
 0 & \longrightarrow & \text{III}^1(k_S/k, T, A) & \longrightarrow & H^1(k_S/k, A) & \xrightarrow{\text{res}^1} & \prod_{\mathfrak{p} \in T(K)} H^1(K_{\mathfrak{p}}, A) \\
 & & \uparrow \text{res} & & \uparrow \text{res} & & \uparrow \text{res} \\
 0 & \longrightarrow & \text{III}^1(k_S, T, A) & \longrightarrow & H^1(k_S, A) & \xrightarrow{\text{res}^1} & \prod_{\mathfrak{p} \in T} H^1(k_{\mathfrak{p}}, A) \\
 & & \uparrow \text{inf} & & \uparrow \text{inf} & & \uparrow \text{inf} \\
 0 & \longrightarrow & \text{III}^1(K/k, T, A) & \longrightarrow & H^1(K/k, A) & \xrightarrow{\text{res}^1} & \prod_{\mathfrak{p} \in T} H^1(K_{\mathfrak{p}}/k_{\mathfrak{p}}, A) \\
 & & \uparrow & & \uparrow & & \uparrow \\
 & & 0 & & 0 & & 0
 \end{array}$$

where $T(K)$ denotes primes of K above those primes in T . By assumption, we have that

$$\delta_K(T) = \delta_k(T \cap \text{cs}(K/k))[K : k] > \frac{1}{p}.$$

By (i), $\text{III}^1(k_S/K, T, A) = 0$ so

$$\text{III}^1(k_S, T, A) \cong \text{III}^1(K/k, T, A).$$

Proposition 1 gives $H^1(K/k, \mu_{p^r}) = 0$ and thus $\text{III}^1(K/k, T, A) = \text{III}^1(k_S, T, A) = 0$ unless we are in the *special case* $(k, 2^r)$ where $H^1(k(\mu_{2^r})/k, \mu_{2^r}) \cong \mathbb{Z}/2\mathbb{Z}$. In this case, if there exists a prime $\mathfrak{p} \in T$ which does not split in $k(\mu_{2^r})$, then

$$G = \text{Gal}(k(\mu_{2^r})/k) = G_{\mathfrak{p}} = \text{Gal}(k_{\mathfrak{p}}(\mu_{2^r})/k_{\mathfrak{p}}).$$

It follows that one of the components of the map res^1 is the identity and must have trivial kernel. This forces $\text{III}^1(k(\mu_{2^r})/k, T, A) = 0$ as well.

If $\mathfrak{p}$ splits in $k(\mu_{2^r})$, then we have that $G_{\mathfrak{p}} \subsetneq G$; let $Z_{\mathfrak{p}}$ denote the fixed field of $G_{\mathfrak{p}}$ (decomposition field of $\mathfrak{p}$). Then the kernel of the map

$$\text{res}_{\mathfrak{p}} : \mathbb{Z}/2\mathbb{Z} \cong H^1(k(\mu_{2^r})/k, \mu_{2^r}) \rightarrow H^1(k_{\mathfrak{p}}(\mu_{2^r})/k_{\mathfrak{p}}, \mu_{2^r})$$

is $H^1(Z_{\mathfrak{p}}/k, \mu_{2^r}^{G_{\mathfrak{p}}})$ from inflation-restriction. However, we have $\mu_{2^r}^{G_{\mathfrak{p}}} = \mu_2$ or $Z_{\mathfrak{p}} = k(\mu_{2^t})$ for some $t \geq 2$, so $\ker(\text{res}_{\mathfrak{p}}) = \mathbb{Z}/2\mathbb{Z}$.

To prove (iii), let p be prime and set p^s to be the maximal p -power dividing $|\text{Gal}(K/k)|$ (where $K = k(A)$). By assumption, there exists $\mathfrak{p} \in T$ such that $p^s \mid |G_{\mathfrak{p}}|$ so $G_{\mathfrak{p}}$ contains a Sylow- p subgroup $G^{(p)}$ of $G = \text{Gal}(K/k)$. The following proposition is proposition 1.6.10 in [2]:

Proposition 3

Let A be a discrete G -module and let $G^{(p)}$ be a p -Sylow subgroup of G . For any Abelian group M , let $M(p)$ denote the p -primary part of M . Assume that G is finite or that $n \geq 1$. Then the morphism

$$\text{res} : \hat{H}^n(G, A)(p) \rightarrow \hat{H}^n(G^{(p)}, A)$$

is injective and if $G^{(p)}$ is open in G , the morphism

$$\text{cor} : \hat{H}^n(G^{(p)}, A) \rightarrow \hat{H}^n(G, A)(p)$$

is surjective.

Since $H^1(G, A)$ is a torsion group, we have that $H^1(G, A) = \bigoplus_p H^1(G, A)(p)$. Combining this with the natural injection $\bigoplus_I G_i \rightarrow \prod_I G_i$ and **proposition 3** gives that the map

$$H^1(G, A) \rightarrow \prod_p H^1(G^{(p)}, A)$$

is injective, so $\text{III}^1(K/k, T, A) = 0$. Chasing around the diagram used in the proof of (ii) shows that $\text{III}^1(k_S, T, A) = 0$. ■

With this, we can obtain the following generalization of **theorem 1**.

Theorem 3

Let k be a global field and let T be a set of primes with $\delta(T) = 1$. Then:

1. The canonical map

$$k^\times / k^{\times m} \rightarrow \prod_{\mathfrak{p}} k_{\mathfrak{p}}^\times / (k_{\mathfrak{p}}^\times)^m$$

is injective, except when $m = 2^r m'$, m' is odd, $r \geq 3$, $k(\mu_{2^r})/k$ is not cyclic, k is a number field, and all primes $\mathfrak{p}$ dividing 2 in k split in $k(\mu_{2^r})$.

2. For any number field k , If α is a $2m^{\text{th}}$ power in $k_{\mathfrak{p}}$ for all $\mathfrak{p} \in T$, then α is an m^{th} power in k .

Proof. Part (1) will be a consequence of part (2) of **theorem 2**. Begin with the Kummer short exact sequences

$$1 \longrightarrow \mu_m \longrightarrow \bar{k}^\times \xrightarrow{\cdot m} \bar{k}^\times \longrightarrow 1$$

$$1 \longrightarrow \mu_m \longrightarrow \bar{k}_{\mathfrak{p}}^\times \xrightarrow{\cdot m} \bar{k}_{\mathfrak{p}}^\times \longrightarrow 1$$

Taking cohomology with respect to G_k and $G_{\mathfrak{p}}$ and applying Hilbert's theorem 90, we obtain the following commutative diagram with exact rows.

$$\begin{array}{ccccccc} k^\times & \xrightarrow{\cdot m} & k^\times & \longrightarrow & H^1(G_k, \mu_m) & \longrightarrow & 1 \\ \downarrow & & \downarrow & & \downarrow \text{res} & & \\ k_{\mathfrak{p}}^\times \cap \bar{k} & \xrightarrow{\cdot m} & k_{\mathfrak{p}}^\times \cap \bar{k} & \longrightarrow & H^1(G_{\mathfrak{p}}, \mu_m) & \longrightarrow & 1 \\ \downarrow & & \downarrow & & \downarrow = & & \\ k_{\mathfrak{p}}^\times & \xrightarrow{\cdot m} & k_{\mathfrak{p}}^\times & \longrightarrow & H^1(G_{\mathfrak{p}}, \mu_m) & \longrightarrow & 1 \end{array}$$

The upshot of this diagram is that the kernel of the map from $k^\times / k^{\times m}$ to the product of the local analogues $k_{\mathfrak{p}}^\times / k_{\mathfrak{p}}^{\times m}$ is precisely $\text{III}^1(k_S, T, \mu_m)$, where S is the set of all primes of k . We see that

$$\delta(\text{cs}(k(\mu_{p_i}^{r_i})/k) \cap T) = \delta(\text{cs}(k(\mu_{p_i}^{r_i})/k)) = \frac{1}{[k(\mu_{p_i}^{r_i}) : k]} > \frac{1}{p_i [k(\mu_{p_i}^{r_i}) : k]}$$

by Chebotarev density, so the conditions to apply **theorem 2** part (2) apply and we obtain the result.

Now we work towards part (2) of the theorem. When we are not in the case where $\text{III}^1(k_S, T, \mu_m) = \mathbf{Z}/2\mathbf{Z}$, the theorem follows from part (1), so we may assume that we are in that case. We may also assume that m is a power of 2, as a number $\alpha \in k^\times$ is an $2^r m'$ th power (with $\gcd(2, m') = 1$) if and only if it is a 2^r th power and an m' th power.

Define $K = k(\mu_{2^{r+1}})$. By inflation-restriction, we have the commutative diagram with an exact column:

$$\begin{array}{ccc}
K^\times / K^{\times 2^{r+1}} & \xrightarrow{\sim} & H^1(G_K, \mu_{2^{r+1}}) \\
\uparrow \text{can} & & \uparrow \text{res} \\
k^\times / k^{\times 2^{r+1}} & \xrightarrow{\sim} & H^1(k, \mu_{2^{r+1}}) \\
& & \uparrow \text{inf} \\
& & H^1(K/k, \mu_{2^{r+1}})
\end{array}$$

Let $\alpha \in k^\times$ be a $2m$ th power in $k_{\mathfrak{p}}^\times$. Let $\mathfrak{p}$ be any prime of K lying over $\mathfrak{p}$. Since $K_{\mathfrak{p}}$ is an extension of $k_{\mathfrak{p}}$, it follows that α is a $2m$ th power in $K^\times$. By exactness of the right column, this implies that α is in the image of $H^1(K/k, \mu_{2^{r+1}}) = \mathbf{Z}/2\mathbf{Z}$, so $\alpha^2 \in k^{\times 2^m}$. If we let $\alpha^2 = \beta^{2^m}$ with $\beta \in k^\times$, we obtain $\alpha = \beta^m$ or $\alpha = -\beta^m$.

We would like to show that we are in the first of these two cases. If $\alpha = -\beta^m$, then $-\alpha$ is an m th power in $k_{\mathfrak{p}}$ for all $\mathfrak{p} \in T$. Since both α is also an m th power in $k_{\mathfrak{p}}$, it follows that -1 is an m th power in $k_{\mathfrak{p}}$ for all $\mathfrak{p} \in T$. Since we are in the special case, we know that $k(\mu_{2^r})/k$ is not cyclic, so in particular complex conjugation is in $\text{Gal}(k(\mu_{2^r})/k)$ and therefore k is real so $\sqrt{-1} \notin k$. Since -1 is a square in $k_{\mathfrak{p}}$ for $\mathfrak{p} \in T$, it follows that $\mathfrak{p}$ does not split in $k(\sqrt{-1})/k$ for all $\mathfrak{p} \in T$. Since $\delta(T) = 1 > 1/2$, this is a contradiction of Chebotarev density. ■

Remark:

Note that we did not use the full power of our assumptions, in particular the $\delta(T) = 1$ assumption. We applied this at two points in the proof. First to ensure we satisfied the assumptions of **theorem 2**, and again at the end to obtain the Chebotarev density contradiction. In practice, given a specific S and m , it is possible to work with a smaller $\delta(T)$, as long as you can ensure both those steps are maintained.

We would now like to say something a little more subtle about $k^\times / k^{\times m}$. For some set of primes S , define the Kummer group V_S as

$$V_S(k, m) := \ker \left(k^\times / k^{\times m} \rightarrow \prod_{\mathfrak{p} \in S} k_{\mathfrak{p}}^\times / k_{\mathfrak{p}}^{\times m} \times \prod_{\mathfrak{p} \notin S} k_{\mathfrak{p}}^\times / \mathcal{O}_{\mathfrak{p}}^\times k_{\mathfrak{p}}^{\times m} \right).$$

Where we set $\mathcal{O}_{\mathfrak{p}}^\times = k_{\mathfrak{p}}^\times$ if $\mathfrak{p}$ is infinite. Note that for finite primes we have the canonical isomorphism induced by the valuation $v_{\mathfrak{p}} : k_{\mathfrak{p}}^\times / \mathcal{O}_{\mathfrak{p}}^\times k_{\mathfrak{p}}^{\times m} \rightarrow \mathbf{Z}/m\mathbf{Z}$.

For some motivation for V_S , recall **theorem 1**, and note that we only actually used the valuation structure in $k_{\mathfrak{p}}$ (except for $\mathfrak{p} = \infty$), so it turns out that we had actually proven the much stronger fact that $V_{\{\infty\}}(\mathbf{Q}, 2) = 0$. In general, V_S tells us how much information we can obtain from just the value group of $k_{\mathfrak{p}}$.

Theorem 4

Let k be a global field, m a natural number coprime to $\text{char}(k)$, and S a set of places of k

1. If $m \in \mathbf{N}(S)$ and S contains all Archimedean primes, we have

$$\text{III}^1(k_S, S, \mu_m) = V_S(k, m)$$

2. The group $V_S(k, m)$ is finite.
3. If $\delta(S) = 1$ and we are not in the special case (k, m, S) , we have $V_S(k, m) = 0$

Proof. We begin with part (1). Recall the Kummer exact sequence

$$1 \longrightarrow \mu_m \longrightarrow k_S^\times \longrightarrow (k_S^\times)^m \longrightarrow 1$$

Taking Cohomology and applying the long exact sequence as usual, we obtain a canonical isomorphism

$$(k_S^{\times m} \cap k^\times / k^{\times m}) \simeq H^1(G_S(k), \mu_m)$$

Working as we did in the proof of **theorem 3**, we obtain

$$\text{III}^1(k_S, S, \mu_m) \simeq \ker \left((k_S^{\times m} \cap k^\times) / k^{\times m} \rightarrow \prod_{\mathfrak{p} \in S} k_{\mathfrak{p}}^{\times} / k_{\mathfrak{p}}^{\times m} \right)$$

Note that this identifies $\text{III}^1(k_S, S, \mu_m)$ as a subset of $k^\times / k^{\times m}$, just like $V_S(k, m)$. We will show that these two kernels are literally equal as sets.

Let $\alpha \in \text{III}^1(k_S, S, \mu_m)$. Then $\alpha = \beta^m$ for some $\beta \in k_S$. Since $k(\alpha^{1/m}) \subseteq k_S$ is unramified outside S , we see that in particular the local extensions $k_{\mathfrak{p}}(\alpha^{1/m}) / k_{\mathfrak{p}}$ is unramified. But this can only be unramified if $m | v_{\mathfrak{p}}(\alpha)$, so it follows that $\alpha \in \mathcal{O}_{\mathfrak{p}}^{\times} k_{\mathfrak{p}}^{\times m}$ for all $\mathfrak{p} \notin S$, so $\alpha \in V_S(k, m)$. Note that it was only valid to work with the valuations because all Archimedean primes are in S .

On the other hand, let $\alpha \in V_S(k, m)$. Then for all $\mathfrak{p} \notin S$, we have $m | v_{\mathfrak{p}}(\alpha)$ and $\mathfrak{p} \nmid m$. (The second follows from the fact that $m \in \mathbf{N}(S)$.) We will argue that the extension $k(\alpha^{1/m}) / k$ is unramified outside S , and we factor this into a lemma because it amazingly comes up 3 chapters later in these notes.

Lemma 4

Let k be a global field, $\alpha \in k$ and $\text{char } k \nmid m$. Let $\mathfrak{p}$ be a prime of k . Suppose $m | v_{\mathfrak{p}}(\alpha)$ and $v_{\mathfrak{p}}(m) = 0$. Then $k(\alpha^{1/m}) / k$ is unramified at $\mathfrak{p}$

Proof. We will show that the local extension $k_{\mathfrak{p}}(\alpha^{1/m}) / k_{\mathfrak{p}}$ is unramified by showing that its discriminant Δ obeys $v_{\mathfrak{p}}(\Delta) = 0$. Define $\alpha' = \alpha / \pi^{v_{\mathfrak{p}}(\alpha)}$ where π is a uniformizer in $k_{\mathfrak{p}}$. Let $\alpha = \gamma^a$ where $a | m$ maximal and $ab = m$, so that the minimal polynomial of α is $x^b - \gamma$. Then the discriminant of $k_{\mathfrak{p}}(\alpha^{1/m}) / k_{\mathfrak{p}}$ (which is just $k_{\mathfrak{p}}(\gamma^{1/b}) / k_{\mathfrak{p}}$) is $\Delta = N_{k_{\mathfrak{p}}(\gamma^{1/b}) / k_{\mathfrak{p}}} (f'(\gamma^{1/b}))$, where $f = x^b - \gamma$ is the minimal polynomial of $\gamma^{1/b}$. We have

$$\Delta = N_{k_{\mathfrak{p}}(\gamma^{1/b}) / k_{\mathfrak{p}}} (f'(\gamma^{1/b})) = N_{k_{\mathfrak{p}}(\gamma^{1/b}) / k_{\mathfrak{p}}} (b\gamma^{(b-1)/b}) = (-1)^b b \gamma^{b-1}.$$

But since $b | m$ and $v_{\mathfrak{p}}(m) = 0$, we have $v_{\mathfrak{p}}(\Delta) = 0 + (b-1)v_{\mathfrak{p}}(\gamma) = 0$, so the extension is unramified. ■

Since $k_{\mathfrak{p}}(\alpha^{1/m})$ is unramified for all $\mathfrak{p} \notin S$, we have that $\alpha^{1/m} \in k_S$, so $\alpha \in k_S^{\times m} \cap k^\times$ and $V_S(k, m) \subseteq \text{III}^1(k_S, S, \mu_m)$.

For part (2), we first use theorem 8.6.4 of [2] which tells us that III is finite for finite $G_S(k)$ -modules A with $|A| \in \mathbf{N}(S)$, so when S, m satisfy the conditions of part (1) we obtain part (2). For the general version of part (2), let S be arbitrary and T the set of all primes dividing $m\infty$.

$$0 \longrightarrow V_{S \cup T}(k, m) \longrightarrow V_S(k, m) \longrightarrow \bigoplus_{\mathfrak{p} \in T-S} \mathcal{O}_{\mathfrak{p}}^{\times} / (\mathcal{O}_{\mathfrak{p}}^{\times})^m$$

The first map is inclusion of sets. The second map is more interesting. If α is in the kernel, then $m | v_{\mathfrak{p}}(\alpha)$ for all $\mathfrak{p}$, so we may associate α with an element of $\mathcal{O}_{\mathfrak{p}}^{\times}$ by multiplying by an element of $k^{\times m}$.

Exactness at $V_{S \cup T}(k, m)$ follows from the fact that we are mapping a kernel of a map to the kernel of the same map but to a quotient of the original codomain, so it is injective. For exactness at $V_S(k, m)$, suppose α is in the kernel of the second map, then at $\mathfrak{p} \in T - S$ it can be written as $u^m \pi_{\mathfrak{p}}^{ma}$ for some $a \in \mathbf{Z}$ and $u \in \mathcal{O}_{\mathfrak{p}}^{\times}$, but then α is an m^{th} power in $k_{\mathfrak{p}}^{\times}$, for all $\mathfrak{p} \in T - S$, so it is also in $V_{S \cup T}(k, m)$.

Since the first and third entries of this sequence are finite (the first by the special case we already solved, and the second by the structure of $\mathcal{O}_{\mathfrak{p}}^{\times}$ and the fact that T is finite) we obtain that $V_S(k, m)$ is finite.

For part (3), this is a consequence of **theorem 3**, since if $\alpha \in V_S(k, m)$, it follows that $\alpha \in \text{III}^1(k_S, S, \mu_m)$. ■

2.3 Local-to-Global for the Artin–Schreier image

In most cases, results about m th powers for $\text{char } k \nmid m$ have an analogue involving the *Artin–Schreier image*. Define the Artin–Schreier map $\wp : k \rightarrow k$ by $\wp(\alpha) = \alpha^p - \alpha$. Then we have the following result of (unsurprisingly) Artin and Schreier.

Proposition 4

Let k be a field with characteristic p and let G_k be the absolute Galois group $\text{Gal}(\bar{k}/k)$. Then there is a canonical isomorphism

$$k/\wp(k) = H^1(G_k, \mathbb{F}_p) = \text{Hom}(G_k, \mathbb{Z}/p\mathbb{Z})$$

This makes a very clear analogy to Kummer theory, where $k^\times/k^{\times m}$ is isomorphic to $H^1(G_k, \mu_m)$.

Proof of proposition 4. Consider the following short exact sequence involving the additive groups of fields:

$$0 \longrightarrow \mathbb{F}_p \longrightarrow \bar{k} \xrightarrow{\wp} \bar{k} \longrightarrow 0$$

Exercise 2

Verify that $\wp$ is a homomorphism of abelian groups and that the above sequence is exact.

Applying cohomology and using the fact that the additive groups of fields are cohomologically trivial (see 6.1.1 in [2]), we obtain the short exact sequence

$$0 \longrightarrow \mathbb{F}_p \longrightarrow k \xrightarrow{\wp} k \longrightarrow H^1(G_k, \mathbb{F}_p) \longrightarrow H^1(G, \bar{k}) = 0$$

looking at exactness at the second instance of $\bar{k}$ and $H^1(G_k, \mathbb{F}_p)$, we obtain the first canonical isomorphism in the proposition. The second follows from using an explicit description of H^1 by twisted homomorphisms and the fact that G_k acts trivially on $\mathbb{F}_p$. ■

Using this proposition, we can motivate (and prove!) the characteristic p analogue of **theorem 3**

Theorem 5

Let S be a set of all primes of a characteristic p field k , and let $T \subseteq S$ have $\delta(T) > 1/p$. Then the canonical map

$$k/\wp(k) \rightarrow \prod_{p \in T} k_p/\wp(k_p)$$

is injective.

Proof. Using **proposition 4** and working as we did in the beginning of the proof of **theorem 3**, we obtain a canonical isomorphism

$$\ker \left(k/\wp(k) \rightarrow \prod_{p \in T} k_p/\wp(k_p) \right) = \text{III}^1(k_S, T, \mathbb{F}_p).$$

By **theorem 2**, the second group is trivial and we obtain injectivity. ■

2.4 Local-to-Global for simple G -modules

We now turn our attention to simple G -modules and prove local-to-global for many cases. To build up to this result (and others) we will use the following lemma

Lemma 5

Let G be a profinite group and let A be a finite simple G module. Suppose there is a normal subgroup $N \subseteq G$ of order coprime to $|A|$ which acts nontrivially on A . Then

$$H^i(G, A) = 0 \quad \text{when } i \geq 0$$

Proof. For $i = 0$, we see that G acts nontrivially on A , so $A^G \subset A$, but since A is simple this implies $A^G = 1$ and therefore $H^0(G, A) = 0$.

Since the order of N is coprime to $|A|$, we obtain that the cohomology groups $H^i(N, A)$ vanish for $i > 0$. This means the inflation-restriction exact sequence is valid for i th cohomology for all $i > 0$. Applying this, we get

$$0 \longrightarrow H^i(G/N, A^N) \longrightarrow H^i(G, A) \longrightarrow H^i(N, A)^{G/N} = 0$$

so we obtain isomorphisms between $H^i(G/N, A^N)$ and $H^i(G, A)$. However, $H^i(G/N, A^N)$ vanishes since $A^N \subset A$ must be trivial, so we obtain $H^i(G, A) = 0$. ■

The following proposition will also be useful

Proposition 5

Let A be a simple G_k module, let $K = k(A)$ be a solvable nontrivial extension of k . Then

$$H^i(K/k, A) = 0$$

Before proving this, we discuss some properties of finite simple G -modules.

Lemma 6

Let A be a finite simple G -module. Then A is an $\mathbb{F}_p$ vector space for some p .

Proof. Let $[p] : A \rightarrow A$ be the multiplication-by- p map. This is a group homomorphism, and moreover it is G -equivariant because $\mathbb{Z}$ is in the center of $\mathbb{Z}[G]$. The kernel of this is clearly a p -torsion abelian group and therefore an $\mathbb{F}_p$ vector space. Since A is simple, it follows that $\ker([p]) = 0$ or $\ker([p]) = A$, and the first cannot be the case for all primes unless $A = 0$. ■

Lemma 7

Let A be a p^n -torsion G -module for a p -group G . If $A^G = 0$, then $A = 0$.

Proof. By orbit-stabilizer, the orbits Ga for some $a \in A$ are of p -power order. Since A^G is simply the set of things with trivial orbit, we obtain $1 = |A^G| \equiv |A| \pmod{p}$, but A is p^n -torsion so we only have $|A| \equiv 1 \pmod{p}$ if $A = 0$. ■

Proof of proposition 5. Since $G = \text{Gal}(K/k)$ is solvable, it has an abelian normal subgroup N . Let N_p be a Sylow- p subgroup, and we claim that N_p is trivial. Since A is simple, $A^{N_p} = 0$ or $A^{N_p} = A$. In the first case we obtain $A = 0$ by lemma 7. In the second, we obtain that $\text{Gal}(\bar{k}/k(A)^{N_p})$ acts trivially on A , so by the definition of $k(A)$ we have $N_p = 1$.

Therefore G has a nontrivial normal subgroup N with order prime to p (since $N_p = 1$) which acts nontrivially on A (if the action was trivial we would contradict the definition of $k(A)$ and the nontriviality of N , just like in the previous paragraph.) From here, it follows via lemma 5 that $H^i(G, A) = 0$. ■

We can now prove the following theorem:

Theorem 6

Let k be a global field, S a set of primes of k , and let A be a finite simple $G_S(k)$ -module. Let p be the prime number with $pA = 0$ and let $T \subseteq S$. Then

$$\text{III}^1(k_S, T, A) = 0$$

in the following cases:

- (i) $\text{Gal}(k(A)/k)$ is solvable and $\text{cs}(k(A)/k) \subsetneq T$
- (ii) $p \in \mathbf{N}(S)$, $\text{Gal}(k(A')/k)$ is solvable and $\text{cs}(k(A')/k) \subsetneq T$
- (iii) $p \in \mathbf{N}(S)$ and there are finite Galois extensions $K \subseteq \Omega$ of k inside k_S such that $p \nmid [\Omega : K]$ and
 - $\mu_p \not\subseteq K$ and $k(A') \subseteq K$
 - $\mu_p \subseteq \Omega$ and $\text{cs}(\Omega/k) \subsetneq T$.

Proof. (i) Let $K = k(A)$. If $K \neq k$, then $H^1(K/k, A) = 0$ by [proposition 5](#); if $K = k$, $H^1(K/k, A) = 0$ trivially. We have the following commutative diagram:

$$\begin{array}{ccccccc}
 0 & \longrightarrow & 0 = \text{III}^1(k_S/K, T, A) & \longrightarrow & H^1(k_S/K, A) & \xhookrightarrow{\text{res}^1} & \prod_{p \in T(K)} H^1(K_p, A) \\
 & & \uparrow \text{res} & & \uparrow \text{res} & & \uparrow \text{res} \\
 0 & \longrightarrow & \text{III}^1(k_S, T, A) & \longrightarrow & H^1(k_S/k, A) & \xrightarrow{\text{res}^1} & \prod_{p \in T} H^1(k_p, A) \\
 & & \uparrow \text{inf} & & \uparrow \text{inf} & & \uparrow \text{inf} \\
 0 & \longrightarrow & 0 = \text{III}^1(K/k, T, A) & \longrightarrow & 0 = H^1(K/k, A) & \xrightarrow{\text{res}^1} & \prod_{p \in T} H^1(k_p, A)
 \end{array}$$

As constructed, A is a trivial $\text{Gal}(k_S/K)$ -module; by assumption, $\text{cs}(K/k) \subsetneq T$, so $\delta_k(T \cap \text{cs}(K/k)) = 1/[K : k]$ by Chebotarev Density Theorem. It follows that $\delta_k(T) = \delta_k(T \cap \text{cs}(K/k))[K : k] = 1$, so $\text{III}^1(k_S/K, T, A) = 0$ by (i) of [theorem 2](#). From this we have $\text{III}^1(k_S, T, A) = 0$ as desired.

(ii) Since $p \in \mathbf{N}(S)$, we have that $k(\mu_p)/k$ is unramified outside S and thus $\mu_p \subset \mathcal{O}_S^\times$. Further, since A is a finite simple G -module, we know by [lemma 6](#) that A is p -torsion and so $A' = \text{Hom}(A, \mu_p)$. Moreover, since A is simple, it follows that A' must also be simple.

Now let $K = k(A)$ and $K' = k(A')$. Then $K'(\mu_p) = K(\mu_p)$; from this we have $H^1(K'(\mu_p)/K, A) = H^1(K(\mu_p)/K, A)$. However, $\text{Gal}(K(\mu_p)/K)$ acts trivially on A by construction, so $H^1(K(\mu_p)/K, A) = \text{Hom}(\text{Gal}(K(\mu_p)/K), A) = 0$ as A is an $\mathbf{F}_p$ -vector space. ■

References

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